

Question_9:

Problem Description:

Tina has given a boolean matrix $mat[P][Q]$ of size $P \times Q$ to Laaysa,

She wants to modify it such that if a matrix cell $mat[m][n]$ is 1 (or true) then make all the cells of m _th row and n th column as 1. Can you help Laaysa?

Constraints:

$1 \leq p, q \leq 1000$

Input Format:

First line of the input is the how many rows and columns in Laaysa matrix.

After that, each line will represent each row and Laaysa need to enter numbers.

Output Format:

Print the resultant matrix output in a separate lines.

✓ Logical Test Cases

Test Case 1

INPUT (STDIN)

```
2 2
1 0
0 1
```

EXPECTED OUTPUT

```
1 1
1 1
```

Test Case 2

INPUT (STDIN)

```
3 3
0 0 0
0 1 0
0 0 1
```

EXPECTED OUTPUT

```
0 1 1
1 1 1
1 1 1
```

Code:

```
1  /*
2  Question_9:
3  If any element of the matrix is 1, make its entire row
4  and entire column equal to 1.
5  */
6
7  #include <iostream>
8  #include <vector>
9  using namespace std;
10
11 int main()
12 {
13     int p, q;
14     if (!(cin >> p >> q)) return 0;
15     if (p < 0 || q < 0) return 0;
16
17     vector<vector<int>> mat(p, vector<int>(q));
18     vector<int> row(p, 0);
19     vector<int> col(q, 0);
20
21     for (int i = 0; i < p; ++i)
22     {
23         for (int j = 0; j < q; ++j)
24         {
25             if (!(cin >> mat[i][j])) mat[i][j] = 0;
26             if (mat[i][j] == 1)
27             {
28                 row[i] = 1;
29                 col[j] = 1;
30             }
31         }
```

```
32     }
33     for (int i = 0; i < p; ++i)
34     {
35         for (int j = 0; j < q; ++j)
36         {
37             if (row[i] == 1 || col[j] == 1)
38             {
39                 mat[i][j] = 1;
40             }
41         }
42     }
43     for (int i = 0; i < p; ++i)
44     {
45         for (int j = 0; j < q; ++j)
46         {
47             cout << mat[i][j];
48             if (j < q - 1) cout << " ";
49         }
50         cout << endl;
51     }
52     return 0;
53 }
```

Output:

Test Case_1:

```
2 2
1 0
0 1
1 1
1 1
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
3 3
0 0 0
0 1 0
0 0 1
0 1 1
1 1 1
1 1 1
PS C:\Users\Prabin Yadav\
```